FLOOD RISK & DISASTER MANAGEMENT

INTEGRATED PLATFORM PROGRESS REPORT: FEATURES, DATA AND MODELS

SUBMITTED FOR	PREPARED BY	REPORTING CUT-OFF
Chairman review	Faiza Muzaffar	4 November 2026

EXECUTIVE SUMMARY

One integrated platform now covers the full flood-management chain: **live monitoring** and national GIS screening, **river and upstream forecasting views**, **citizen reporting, alert approval and incident response**, a **provenance-tracked data observatory**, **42 trained and evaluated forecast/risk model configurations**, and a **multi-engine flood simulation studio** with population and infrastructure exposure. Monitoring and response features run on live public data; forecasting and simulation outputs are functional and in pilot status pending more observed gauge and survey data.

10 portal workspaces	**55+** backend API endpoints	**126** districts in national risk screening	**28** priority barrages, dams & gauges
3,209 eligible Marala daily discharge readings	**42** forecast & risk model configurations	**4** flood simulation engines	**385** hourly flood frames published

1. OPERATIONS PORTAL AND DASHBOARD live monitoring and command view

- **Operational overview:** KPI cards, command stats deck, environmental panel and live risk panel on one screen.
- **Live data feeds:** Open-Meteo weather and GloFAS river discharge; strict live mode returns errors instead of mock values.
- **Risk engine:** rules-based scoring (*rule_based_v1*) of live inputs per monitoring location; trend detection.
- **Hourly analytics ticker:** rolling hourly rainfall and discharge indicators.
- **10 workspaces:** Overview, GIS, Predictive Risk, Upstream Watch, River Monitoring, Alerts, Incidents, Citizen Reports, Activity, Simulation.
- **Guided portal tour:** step-by-step in-place walkthrough for new users without disabling controls.
- **System status panel:** provider health, data mode and model state visible to operators.
- **Backend:** FastAPI with 55+ REST endpoints, OpenAPI docs, SQLite now and PostgreSQL-ready; emergency-resource and automation-context feeds.

2. GIS OPERATIONS AND MAPPING national hazard screening

- **National risk layer:** 126 ADM2 districts, gzip GeoJSON, viewport (bbox) filtering and server-side caching.
- **Map layers:** rainfall heatmap raster, rivers, glaciers, HydroBASINS catchments, region outlines, LoC reference.
- **Kashmir context:** 10 AJ&K + 22 context districts; boundary-source selector (district-aligned / previous outline).
- **Fast loading:** content-hashed .geojson.gz assets, browser Cache Storage, boundary-first paint, background layer warm-up.
- **Coordinate tools:** lat/long entry, click-to-pick, locate marker, deselect, Mercator-range validation (±85.05°).
- **Catchment study mode:** isolates the pinned district and sub-basin; clips rivers, gauges and rainfall to it.
- **Terrain studio:** 3D relief, 58° bird's-eye camera, 1-2x exaggeration; presets for Tarbela, Mangla, Marala, Taunsa.
- **Satellite water extent:** Sentinel-1 VV/VH + Copernicus DEM + JRC water, 256x256 px at 30 m, U-Net mask overlay; GeoTIFF upload (≤32 MB).
- **Google Flood Hub & water intelligence:** external flood-forecast and water layers in the same map.
- **Location details:** district, catchment, glacier and screening polygons for any picked point.

3. RIVER MONITORING AND FORECAST VIEWS barrages, upstream stations, outlooks

- **Priority network:** 15 barrages/headworks, 6 upstream stations, 3 dams, 4 confluences; per-category toggles and symbols.
- **Station hover cards:** 120 ms hover intent, click-to-pin, keyboard cycling, 48 h sparkline, exact 3 h / 24 h change, stale flags.
- **Station details:** readings table, hydrograph and upstream network; official FFD history endpoint.
- **Discharge forecast window:** FFD observations + modelled history + 5 forecast days, selectable legend, date inspector, CSV export.
- **Upstream Source Flood Watch:** upper-basin discharge forecasts with flood-threshold lines, station location map, cached refresh.
- **River monitoring table:** barrage and headworks readings with manual refresh and FFD bulletin ingestion.
- **Seasonal Basin Risk:** basin-level seasonal risk view in Predictive Risk Views.
- **Monthly Marala flow:** monthly dataset and 365-day outlook (16 Oct 2026 - 15 Oct 2027) with CSV download.
- **10-Year Outlook:** scenario maps and charts, flood-spread and population layers, lowest/highest cases, animated tour, inputs and method panel.
- **Experimental hydrograph:** observed versus modelled hydrograph modal and predictive-inputs dialog.

4. ALERTS, INCIDENTS AND CITIZEN REPORTING response workflow

- **Citizen reports:** two-way: read from and write to the database; scrollable live feed.
- **Report location:** map location picker, current-location button and lat/long display.
- **Photo geotags:** EXIF location extracted automatically; empty fields shown as null, never guessed.
- **Report actions:** mark important and close report, with status saved to the database.
- **Early-warning approval:** critical alerts pass an approval step before escalation to higher authorities.
- **Alerts register:** threshold crossings and monitoring alerts, up to 500 recent records.
- **Incident management:** create, update and track incidents with location, status and history.
- **System activity timeline:** automated decisions, monitoring changes and response events in one log.
- **Monitoring scenarios:** station test scenarios with play, save and threshold-alert checks (isolated test mode).
- **Population analysis:** region, admin level and flood source selection; async analysis jobs with map layers.

5. DATA OBSERVATORY AND TRAINING DATASETS provenance, quality, coverage

- **Local observatory:** one-command portal, rebuilds snapshot and re-verifies SHA-256 checksums at launch.
- **Overview:** date/year filters, eligible vs withheld inflow, monthly usability, distributions, annual maxima.
- **Summary view:** 6 aligned panels (inflow, outflow, rainfall, IRSA, PMD); fit-to-screen or scrollable year.
- **Chart tools:** linked crosshair, double-click date lock, Esc unlock, synced per-panel scrollbars, exact-date tooltips.
- **Rainfall comparison:** combined chart plus 4 single-source plots, per-year filter, independent availability chart.
- **Availability calendar:** every day from 2 Jul 1970 to 23 Oct 2026, with review-required markers.
- **Sources and library:** exact URLs, checksums, downloadable CSVs and QC reports.
- **Gap analysis:** nearby-station regression: Khanki 2026 NSE 0.811, Qadirabad 0.777, Trimmu 0.336.

DATASET	SOURCE	COVERAGE / VOLUME	USED FOR
Marala daily discharge	WAPDA, IRSA	Mar 2016 - Dec 2024; 3,209 eligible, 7 withheld	Forecast target
Marala 06:00 gauge	FFD / PMD bulletins	Jul 2024 - Oct 2026; 322 readings	Station forecast
Catchment rainfall	CHIRPS v3; ERA5-Land, CHIRPS v2, PERSIANN-CDR, IMERG V07, NASA POWER	1990 - 2026 daily	Model input, comparison
Meteorology	ERA5-Land: temperature, dewpoint, solar radiation, PET	1980 - 2024 daily	Model input
Snow	ERA5-Land SWE, snow depth, snowmelt; MODIS NDSI cover	1980 - 2024; 2000 - 2019	Model input
Flood history and nearby gauges	PMD peaks, historical search; FFD Khanki, Qadirabad, Trimmu	1970 - 2026	Availability, gap analysis
Pretraining basins	Caravan (global), CAMELS-US (8 basins)	Multi-basin daily	Transfer learning
Terrain and hydrography	FABDEM, WorldCover, HydroRIVERS, HydroBASINS, SoilGrids, 17 FFC reports	Kabul / Nowshera	Flood simulation
Exposure	WorldPop 2025 (100 m), OSM, Microsoft buildings, OCHA COD-AB, UNOSAT 4197	Jhang district	Impact estimates
Live feeds	Open-Meteo, GloFAS, Sentinel-1 RTC, Copernicus DEM, JRC GSW, Global Flood Database	National, on demand	Portal, risk models

6. MODEL TRAINING what was trained, on which data, with which results

- **Target:** Marala daily discharge (m³/s), log-standardised, quantile outputs for uncertainty.
- **Inputs:** 8 daily drivers (CHIRPS rain, temperature, SWE, snow depth, snowmelt, dewpoint, solar radiation, PET) + calendar.
- **Look-back and lead:** 365-day hindcast; 10, 15 and 30-day forecast horizons.
- **Chronological split:** train 2016-2020, validate 2021, test 2022-2024; observed days only, no gap filling.
- **Flood threshold:** training Q95 = 2,495 m³/s; 51 flood days out of 1,075 test targets.
- **Forecast stand-in:** ERA5-Land future weather with lead-dependent noise until archived forecasts are added.

GROUP	RUNS	MODELS TRAINED
10-day horizon	19	Hindcast-forecast LSTM, past-discharge LSTM, 2-input LSTM, Caravan global LSTM (fine-tuned), CAMELS-US 8-basin LSTM, HBV-LSTM, HBV-LightGBM, HBV residual, snow-bucket hybrid, δHBV, elevation-band and catchment-mean LSTM, N-HiTS, N-BEATSx, gradient boosting, LSTM+boosting ensemble, persistence, climatology
15-day horizon	5	Encoder-decoder LSTM (scratch, fine-tuned), discharge-history LSTM, GRRR LSTM, GRRR weighted-fill (rejected)
30-day horizon	12	Weather-only, past-discharge and climatology-residual LSTMs, fine-tuned encoder-decoder, δHBV, calibrated HBV, HBV-LSTM and HBV-LightGBM residual, elevation-band LSTM + control, N-HiTS, N-BEATSx
Portal models	6	Rule-based v1 (operational), XGBoost riverine (Trimmu 24 h, Balloki 48 h), XGBoost on Global Flood Database (ROC-AUC 0.59, gated), Marala gauge ridge/XGBoost (1, 3, 7 days), 365-day seasonal outlook, SAR U-Net water mask
Hydrologic	1	HEC-HMS Kabul routing to Nowshera: validation NSE 0.678, KGE 0.462 (0.469 with Warsak inflows)

DAY-10 TEST (2022-2024)	FLOOD HIT RATE	CSI	FLOOD RMSE (m3/s)	NON-FLOOD NSE
Catchment-mean LSTM	66.7%	0.442	704	0.846
Caravan global LSTM, fine-tuned	64.7%	0.429	707	0.861
Snow-bucket + LSTM hybrid	62.7%	0.348	730	0.767
Hindcast-forecast LSTM	58.8%	0.411	746	0.837
Gradient boosting	31.4%	0.250	811	0.840
Persistence (baseline)	27.5%	0.159	1,184	0.610

Best overall lead-10 NSE: 0.876 (Caravan global LSTM, fine-tuned). Best day-30 flood recall: 74.5% with HBV-LSTM residual hybrid (RMSE 559 m3/s). Marala 06:00 gauge model: 9.5% lower RMSE than persistence at 7 days. Scores use reanalysis weather and are retrospective, not operational forecast accuracy.

7. FLOOD SIMULATION STUDIO scenario to map playback

SCENARIO SET-UP

- **Scenario editor:** dates, domain, mesh, UTC output step, rainfall product and bias correction, soil state, inflows, structures, skill targets.
- **8 experiment families** plus ready examples: Nowshera 2022 and Wetter monsoon +30%; drafts saved, JSON import/export.
- **Case runner:** partial-JSON overrides, server-side validation, one-click Prepare & run, blocked-reason display.
- **Station catalog:** 29 reference stations; Warsak upstream, Kabul-Indus link at Attock downstream.

DATA AND COMPUTE

- **Input registry:** DEM, rivers, catchment, soil, moisture, rainfall, land cover, inflow, validation; licence and SHA-256 stored.
- **Domain prep:** channel snapping within 2 km, native-CRS clipping, coverage and memory checks, hash-based cache; STAC COG import.
- **HEC-RAS 6.6 2D:** full SWE solver, 200 m mesh, 72 h warm-up, 385 hourly frames; zoned-roughness case KGE 0.558.
- **LISFLOOD-FP 8.0.3:** 100 m screening grid, 72 h steady-flow warm-up; HEC-HMS 4.14 hydrology.
- **ANUGA trial (Jhang):** 10,000 cells, 6 h inflow, numerical checks passed.
- **Bring your own model:** engine selector and attach-prepared-model binding by exact input hashes.

RESULTS AND REVIEW

- **Publishing:** GeoTIFF to map overlays with unit, datum, grid and timestamp checks; NSE/KGE vs gauges and observed extent.
- **Playback:** 3D globe, 1x/4x/16x speed, keyboard stepping, depth/WSE/speed legends, hydrograph sync.
- **A/B comparison:** HEC-RAS vs LISFLOOD split slider with nearest-frame matching (±60 s).
- **Provenance and verification views:** data inventory, inflow-improvement and roughness studies, survey status.
- **Portal replays:** storm replay with animated currents and line inspection, flood spread and population bars, cinematic 3D, HEC-RAS confluence/floodplain replays.
- **Maintenance:** WSE restyling, job pruning, failed-job recovery; CI runs lint, build and tests.

8. EXPOSURE AND IMPACT ANALYSIS who and what is in the flood path

- **Study area:** Jhang district (Trimmu) from OCHA COD-AB national release of 160 districts.
- **Population:** WorldPop 2025 constrained grid at 100 m.
- **Infrastructure:** OSM roads, buildings, facilities and settlements clipped to the district.
- **Cross-check:** Microsoft building footprints matched to OSM within 10 m to find mapping gaps.
- **Flood reference:** UNOSAT product 4197 dated VIIRS water extent for historical comparison.
- **Automated pipeline:** download, preprocess, source verification and audit with logs; Google Colab version.
- **Portal link:** exposure regions, analyses and map layers served to the portal by API.
- **Engineering quality:** 54 automated test suites across the platform, plus Playwright browser checks.

9. STATUS AND NEXT STEPS

AREA	STATUS	NEXT STEP
Monitoring, GIS, reporting, alerts	Operational on live public feeds	Hosted multi-user deployment
Forecast models	Trained, evaluated, pilot	Add archived forecasts; connect best model to portal
Flood simulation	Runs end to end, pilot	Calibrate with surveyed channels and flood extents
Observed data	Sparse public gauge records	Request records from WAPDA, FFD/PMD, IRSA, irrigation depts.

DECISION SOUGHT

Approval to request observed river, operating and survey records from partner agencies. These data are the main dependency for moving the platform's forecasts and flood simulations from pilot to operational use.